A Probabilistic Sample Boosting Approach With Adaptive Representation Coefficient Consistency for China Coastal Wetland Land Cover Classification Using GF-5 Hyperspectral Imagery

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***Abstract*— Wetland contains numerous features, and label acquisition is time-consuming, laborious, and inaccurate. Coastal wetland land cover classification with limited labeled training samples has become a significant challenge. In this study, a novel probabilistic ensemble sample selection (ProESS) framework is proposed for coastal wetland land cover classification. First, a sample probabilistic confidence index (SPCI) is proposed, which is defined by probabilistic output of each base classifier. Then, the prediction confidences of unknown samples are obtained by joint probability of ensemble base classifiers, which can select high-quality samples to improve classification performance. However, the low accuracy of base classifiers will affect the confidence of samples selected by SPCI, thus reducing the classification accuracy. Based on this observation, an adaptive representation coefficient consistency learning (AdaRCCL) is proposed to help define SPCI. Finally, a ProESS is constructed through SPCI and AdaRCCL, which can obtain training samples with high confidence from unknown samples. To evaluate the effectiveness of the proposed methods, the three wetland hyperspectral datasets of China, i.e., Yangtze River Delta, Jiangsu Dafeng Natural Reserve, and Yellow River Delta, are used for classification experiments in the article. Experimental results show that the proposed algorithms achieve higher performance and are robust to parameters in comparison to the baseline and the state-of-the-art ensemble algorithms. The extensibility and transferability of the proposed methods are also discussed in this article. Better results on multiple machine learning models with new samples show the extensibility of SPCI and ProESS. The great performance on Botswana dataset also demonstrates the transferability of the proposed methods.**

***Index Terms*— Classification, ensemble learning, hyperspectral, small-sample problem, wetlands.**

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I. Introduction

WETLANDS are essential to the ecosystem of the Earth and have significant ecological, economic, and social values. Coastal wetland land cover classification refers to the classification of wetland landforms in coastal areas, usually including water bodies, marshes, beaches, grasslands, forests, and other types. The purpose of coastal wetland land cover classification is to better understand the structure and function of wetlands by classifying different types of wetlands and to provide scientific basis for the protection, management, and sustainable utilization of wetlands [1], [2], [3]. Therefore, accurate coastal wetland land cover classification has important scientific significance and practical value for wetland management and protection [4], [5]. Hyperspectral data have abundant ground object information and rich hidden features in hundreds of contiguous narrow spectral bands, which can be used as a reliable technical means for accurately monitoring wetlands [6]. A large number of training samples are required to finely classify wetland species by hyperspectral remote sensing technology [7], [8]. However, due to the large variety of wetland species and unique geographical location, it is difficult to obtain enough usable labeled sample [9]. How to deal with the problem of insufficient labeled samples is a crucial problem to overcome in wetland hyperspectral image (HSI) classification. Therefore, the research on wetland HSI algorithm under small-sample size has important theoretical significance and application value.

Few-shot learning (FSL) is a machine learning method that aims to build accurate classification or regression models with less training data. FSL can overcome the limitation of available samples in wetland HSI classification. One of the well-known categories in FSL is data-driven strategy [10]. The introduction of prior knowledge serves the purpose of augmenting the number of training samples in these methods. Several deep learning [11], [12] methods are widely utilized in FSL. Within the framework of the depthwise feature interaction [13] model, meaningful feature interactions can be discovered within limited samples through the application of meta-learning or transfer learning methods when combined with FSL. In the context of the asymmetric feature fusion model [14], FSL can be employed to optimize the process

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of feature fusion, enabling the model to concentrate more on the key features that aid in classification, despite these features possibly originating from scarce samples. Regarding the cross-scale mixing attention model [15], an attention mechanism can be designed to prioritize important scale information on limited samples. This method is capable of assisting the model in the quick and accurate identification and focuses on key information in new remote sensing images. In the setting of adversarial complementary learning [16], the generalization ability of the model can be enhanced when FSL is incorporated. For instance, a strategy can be designed that allows the model to learn an adversarial strategy in FSL to strengthen its capacity to distinguish between different categories. Additionally, through the use of complementary learning, a deeper understanding and processing of new remote sensing images can be achieved by the model as it learns from various perspectives on limited samples. The existing data-driven FSL method can be drawn into the following three categories.

Data augmentation by hand-crafted rules is a classic method used in early research to solve the small-sample problem in image classification. This method mainly uses hand-crafted rules, such as flipping [17], scaling [18], rotation [19], and translation [20], to preprocess the image for data enhancement and amplification. The invariance advantages of the image can be well used to mine the domain information for classification in these methods. Data augmentation methods have been widely used in the field of HSI classification [21], which are regarded as a data preprocessing process in HSI classification [22], [23], [24]. Since this type of method mainly deals with the invariance of specific images, the extensibility and scalability of the algorithm are very low, and the same variation method is challenging to apply to most classification problems. Moreover, domain information is difficult to extract due to the complex types of wetland features and fragmented samples. Therefore, this method is unsuitable for solving the small sample problem of wetland HSI classification.

Another commonly used data-driven FSL method is the diversity-driven method based on ensemble learning [25], [26], which is summarized as follows. It augments the diversity of training data by resampling them into different data spaces or feature spaces. Thus, the new samples are generated by subdatasets or subfeature sets. The most famous methods are the bootstrap [27], [28], [29] and random subspace method [30], [31], which are used to add data and feature attributes variations to the training samples, respectively. Recently, such FSL methods have also been widely used for HSI classification. Su et al. [32] proposed a TCRC-bagging model with sample diversity based on local tangent space manifolds and Bootstrap ideas, which considered the neighborhood pixel locality of the target pixel in the tangent space to effectively improve the classification performance of HSIs. Su et al. [33] also proposed a random subspace K-nearest neighbor class collaborative representation (CR) ensemble algorithm using the idea of random subspace and K-nearest neighbors. This method solved the negative impact of the training subset that did not consider irrelevant categories in the CR regularization algorithm and effectively solved the high-dimensionality problems for hyperspectral data. However, the diversity of data and features will be insignificant if using a small number of initial training samples. Moreover, one of the primary advantages of hyperspectral data is the rich band feature information. The feature diversity of the data will be increased, while the band information may be lost simultaneously using the random subspace method. For wetland HSI, another problem is that some classes will be lost in the subdataset using the bootstrap method. It will significantly reduce the classification accuracy in further steps.

The last category of the FSL strategy augments training samples by selecting samples with high confidence from the testing dataset or unlabeled dataset [34], [35]. In image classification problems, a large amount of testing data and unlabeled data with rich image information are not well utilized. Therefore, the main idea of this FSL method is to select helpful samples from the unlabeled samples to improve classification performance through some machine learning methods. The major issue of these methods is how to define a reliable index to add high-confidence samples into training samples. The self-training [36], [37], [38], co-training [39], and tri-training [40], [41], [42] are the most common FSL methods used for HSI. The basic idea of these methods is to classify unlabeled samples with one or multiple classifiers. Then, some samples with high confidence are selected from the unlabeled sampling for labeling, and pseudolabeled samples are used for further training. However, all three strategies rely too much on the confidence of the results of the initial model. Due to the low accuracy of initial training samples, the results of most classifiers may be wrong. Meanwhile, a large number of noisy samples will be generated when selecting unlabeled samples, reducing the classification accuracy.

According to the above analysis, the data-driven FSL method has the following problems. A specific strategy is needed to design different datasets using a traditional hand-craft ruled method, which does not have generalization ability. The diversity of data and feature is increased to achieve the effect of data enhancement and amplification by resampling based on the bootstrap and random subspace methods. However, due to the complexity of wetland species and the characteristics of hyperspectral data, these two methods are also unsuitable for wetland HSI. Meanwhile, the problem of low initial classification accuracy caused by small samples leads to transfer errors in subsequent iterative data augment.

Therefore, a novel probabilistic ensemble sample selection (ProESS) framework is proposed based on the sample probabilistic confidence index (SPCI) and adaptive representation coefficient consistency learning (AdaRCCL) method for Chinese coastal wetland land cover classification. The goal of the SPCI algorithm is to construct an SPCI by the probabilistic output of the base classifier to make full use of unknown samples with high confidence to assist the output of the base classifier classification. Then, the AdaRCCL algorithm is designed to improve the reliability of the sample confidence based on the assumption that the representation coefficient

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is consistent with each base classifier. The consistency of base classifiers is crucial in ensemble learning [43]. Consistency is referred to as the similarity in error patterns among the base classifiers, enabling the ensemble model to benefit from their combined predictions. Moreover, consistency can be enhanced by utilizing different training data, adjusting algorithm parameters, or selecting feature subsets, thereby further enhancing the performance of the ensemble model. Finally, the AdaRCCL is used to help SPCI construct a novel ProESS framework, which can obtain more reliable samples. The proposed methods can utilize the rich information of unknown samples by a probabilistic sample boosting approach and solve the small sample size problem in wetland HSI classification. The specific aims and contributions in the article can be drawn as follows.

1) To cope with the limited labeled samples problem in wetland HSI classification, a small-size ensemble sample selection method (ProESS) is proposed with SPCI and AdaRCCL. The joint output probability and the pattern consistency of multiple classifiers are considered in this method, which can assist in obtaining the samples with high confidence from unknown samples.
2) A novel ensemble samples selection index (SPCI) is defined to select the samples with higher confidence from unknown samples for training dataset augmentation. The training samples are updated repeatedly according to the reconstruction error obtained from the new training samples until the convergence condition is reached to improve the classification accuracy of the ensemble learning.
3) To better exploit the advantages of unknown samples, an iterative representation coefficient reconstructing strategy (AdaRCCL) is imposed to improve the above sample selection method. The problem of increasing the transfer error of sample amplification due to fewer initial samples or insufficient accuracy of the base classifier is solved by representation coefficient consistency.

The remainder of this article is organized as follows. In Section II, the related works are introduced. In Section III, the proposed SPCI, AdaRCCL, and ProESS are presented. The results of the experiment are demonstrated in Section IV. In Section V, the robustness, extensibility, transferability, and convergence of the proposed algorithms are discussed in detail. Finally, the conclusions of this study are drawn in Section VI.

II. Related Work

A. Collaborative Representation-Based Classification (CRC)

CRC is first proposed for face recognition and later widely used for HSI classification due to its outstanding performance [44]. Similar to sparse representation classification (SRC) [45], the objective is to find the optimal linear representation of the testing samples.

Give a training set $\boldsymbol{X}_{\text{tr}} = \{(\boldsymbol{x}_{\text{tr},i}, \hat{y}_{\text{tr},i})\}$, $i = 1, \ldots, N$; $\boldsymbol{x}_{\text{tr},i} \in \mathcal{X}^d$; $\hat{y}_{\text{tr},i} \in \mathcal{Y}^d$; $|\mathcal{Y}| = K$, and a testing set $\boldsymbol{X}_{\text{te}} = \{(\boldsymbol{x}_{\text{te},j}, \hat{y}_{\text{te},j})\}$, $j = 1, \ldots, M$; $\boldsymbol{x}_{\text{te},j} \in \mathcal{X}^d$; $\hat{y}_{\text{te},j} \in \mathcal{Y}^d$; $|\mathcal{Y}| = K$. The CR coefficient $\boldsymbol{\alpha}$ is obtained as the following formula:

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} \ \|\boldsymbol{X}_{\text{te}} - \boldsymbol{X}_{\text{tr}}\boldsymbol{\alpha}^*\|_2^2 + \lambda\|\boldsymbol{\alpha}^*\|_2^2 \tag{1}$$

where $\|\boldsymbol{X}_{\text{te}} - \boldsymbol{X}_{\text{tr}}\boldsymbol{\alpha}^*\|_2^2$ is the reconstruction error, λ is the regularization parameter, and $\lambda\|\boldsymbol{\alpha}^*\|_2^2$ controls the complexity of $\boldsymbol{\alpha}^*$. By setting the partial derivative to zero, the analytical solution can be easily obtained

$$\boldsymbol{\alpha} = \left(\boldsymbol{X}_{\text{tr}}^T\boldsymbol{X}_{\text{tr}} + \lambda\boldsymbol{I}\right)^{-1}\boldsymbol{X}_{\text{tr}}^T\boldsymbol{X}_{\text{te}} \tag{2}$$

where $\boldsymbol{I}$ denotes the identity matrix. The weights on training samples to reconstruct the testing samples are assigned by $\boldsymbol{\alpha}$.

To solve the problem of linear inseparability of pixels in image classification, kernel collaborative classification (KCRC) is proposed [46]. The basic idea of KCRC is to represent the testing samples through a linear combination of training samples in a kernel space and then reconstructs the testing samples based on the kernel representation coefficients. The optimization problem can be rewritten as

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} \ \|\boldsymbol{\Phi}(\boldsymbol{X}_{\text{te}}) - \boldsymbol{\Phi}(\boldsymbol{X}_{\text{tr}})\boldsymbol{\alpha}^*\|_2^2 + \lambda\|\boldsymbol{\alpha}^*\|_2^2. \tag{3}$$

Similar to (2), the analytical solution can be rewritten as

$$\boldsymbol{\alpha} = (\boldsymbol{\Phi}(\boldsymbol{X}_{\text{tr}})^T\boldsymbol{\Phi}(\boldsymbol{X}_{\text{tr}}) + \lambda\boldsymbol{I})^{-1}\boldsymbol{\Phi}(\boldsymbol{X}_{\text{tr}})^T\boldsymbol{\Phi}(\boldsymbol{X}_{\text{te}}) \tag{4}$$

where $\boldsymbol{\Phi}(\boldsymbol{X}_{\text{te}})$ and $\boldsymbol{\Phi}(\boldsymbol{X}_{\text{tr}})$ represent the testing samples. The training samples are projected into the kernel space using the kernel trick, which is expressed as

$$\kappa(\boldsymbol{x}_{\text{tr},i}, \boldsymbol{x}_{\text{tr},i}) = \boldsymbol{\Phi}\left(\boldsymbol{x}_{\text{tr},i}\right)^T\boldsymbol{\Phi}\left(\boldsymbol{x}_{\text{tr},i}\right) \tag{5}$$

where $\kappa(\cdot,\cdot)$ is a kernel function, e.g., a Gaussian function. Similar to (4), the analytical solution can be rewritten as

$$\boldsymbol{\alpha} = (\kappa(\boldsymbol{X}_{\text{tr}}, \boldsymbol{X}_{\text{tr}}) + \lambda\boldsymbol{I})^{-1}\kappa(\boldsymbol{X}_{\text{tr}}, \boldsymbol{X}_{\text{te}}) \tag{6}$$

where $\kappa(\boldsymbol{X}_{\text{tr}}, \boldsymbol{X}_{\text{tr}})$ is an $N \times N$ Gram matrix and $\kappa(\boldsymbol{X}_{\text{tr}}, \boldsymbol{X}_{\text{te}})$ is equivalent to $\boldsymbol{\Phi}(\boldsymbol{x}_{\text{tr},i})^T\boldsymbol{\Phi}(\boldsymbol{x}_{\text{tr},i})$. The information used for classification is usually the residual corresponding to the kth class

$$\boldsymbol{r}_{\boldsymbol{k}}(y) = \|\boldsymbol{X}_{\text{te},k} - \boldsymbol{X}_{\text{tr},k}\boldsymbol{\alpha}_k\|_2^2 \tag{7}$$

$$\text{class}(y) = \arg\min_k \ \boldsymbol{r}_k(y) \tag{8}$$

where $\boldsymbol{r}_{\boldsymbol{k}}(y)$ is the difference between the testing samples and its reconstructed value, which is obtained by the reconstruction error $\|\boldsymbol{X}_{\text{te},k} - \boldsymbol{X}_{\text{tr},k}\boldsymbol{\alpha}_k\|_2^2$.

B. Confidence Measurement in Ensemble Selection (ES)

ES [47] is a selective ensemble learning approach that aims to find an optimal combination of base classifiers that is better than using the whole ensemble. Therefore, an important issue of this method is to define a reliable classifier confidence index. Based on these assumptions, some confidence measurement methods were proposed. The confidence measurement can be mainly grouped into two categories: supervised methods and unsupervised methods.

On the one hand, the k-nearestNeighbor (KNN) and clustering methods are commonly utilized in supervised methods to define the region of competence (RoC). In dynamic ensemble selection (DES), the RoC is defined as a subset of training

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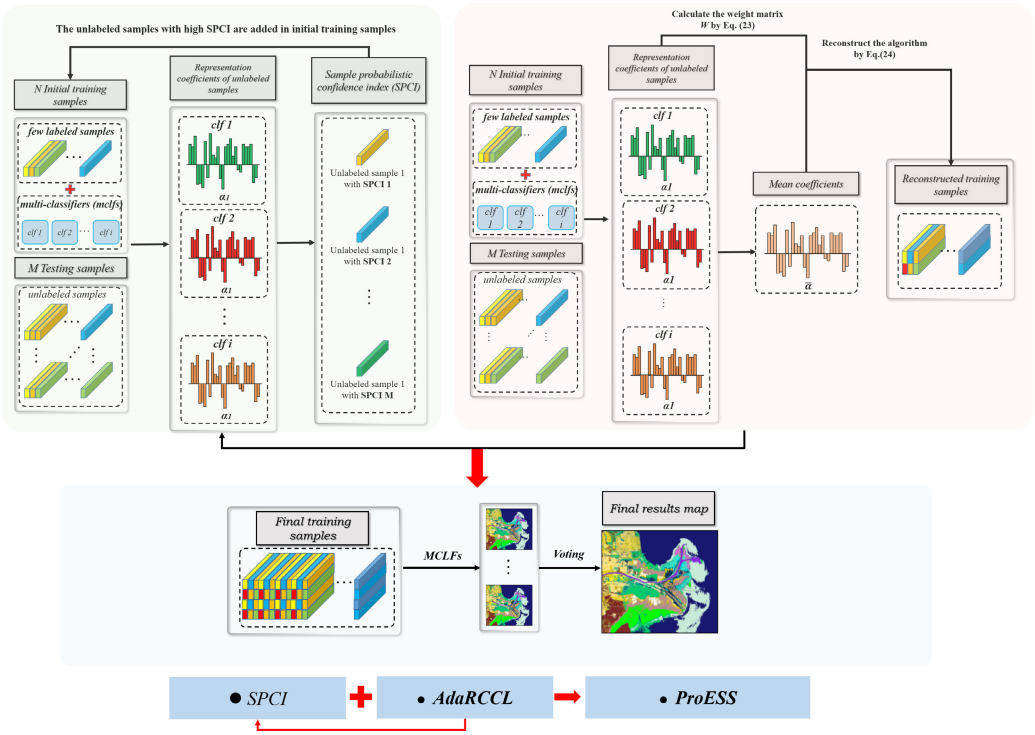


Fig. 1. Flowchart of the proposed methods.

instances that are most similar or closest to a given test instance. Relationships between test samples and training samples are established using distance or similarity measures, resulting in the division of the sample space into distinct regions. Base classifiers that perform well are selected to make predictions for each region, effectively enhancing the ensemble model's classification accuracy and generalization ability. The classification performance of the RoC is used to define the confidence of each base classifier. The classification ability of each base classifier is learned from the RoC-based classifiers, and then, an optimal subset is selected according to classifier confidence. The classic and recent studies on supervised methods are Meta-DES, DES-KNN, and DES-CLUSTER [48].

On the other hand, some optimization-based confidence definition methods are used in unsupervised methods. The selective ensemble is usually formulated as an optimization problem in these approaches. Then, the classifier confidence index can be calculated by solving the optimization problem. The classic and recent studies on supervised methods can be found in [47].

III. Proposed Method

The proposed algorithm is shown in Fig. 1. The algorithm is centered on two key components, namely, sample augmentation and boosting. The traditional multiclassifier sample augmentation approach may potentially result in error propagation among different classifiers, which can lead to a degradation in the classifier's performance. To address this challenge, novel concepts, such as probabilistic reconstruction error and probabilistic confidence index, are introduced. Specifically, the probabilistic reconstruction error considers the relationship between the representation coefficient of testing samples, the initial training data matrix, and the initial testing data matrix and hence provides a more accurate reflection of the classifiers' predictive capability on testing samples. Meanwhile, the probabilistic confidence index is further utilized to enhance the confidence and reliability of the classifiers by conducting probability calculations on the output of each base classifier. These new concepts effectively integrate the output results of different classifiers, mitigate the impact of error propagation, and ultimately improve the efficiency and accuracy of sample augmentation and collaborative learning.

Furthermore, the utilization of Shannon entropy to measure the confidence of the joint output of multiple classifiers can further optimize the efficiency and accuracy of sample augmentation and collaborative learning. Specifically, the uncertainty and randomness of information can be evaluated by Shannon entropy, which is fundamental in assessing the confidence and reliability of the output results of each classifier [49]. By calculating the Shannon entropy of the probability output of each classifier, the analysis of uncertainty and randomness of information can be performed, and the confidence and reliability of the joint output of multiple classifiers can be estimated. Hence, utilizing Shannon entropy to measure the probabilistic confidence index can provide a more precise and dependable assessment of the joint output of multiple classifiers, enhancing the efficiency and accuracy of sample augmentation and collaborative learning.

Building upon the above content, the reconstruction error derived from the classifier pool can be utilized to construct the SPCI, which determines the confidence of testing samples. Specifically, the SPCI is constructed by first computing the

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probabilistic representation index (PRI) of each testing sample using the reconstruction errors of the classifier pool. Then, the confidence of each testing sample is calculated by computing the Shannon entropy of the PRI values for each base classifier. Finally, testing samples with the top-P largest SPCI values are selected to assist in classifying the unknown samples. Specifically, for these labeled testing samples, pseudolabels are assigned to them and they are added to the original training samples. This approach effectively identifies high-confidence testing samples, which are subsequently utilized to improve the classification performance of the system. Although the accuracy of base classifiers can be improved by SPCI, its definition becomes unreliable when the classification performance of initial base classifiers is low. To address this issue and improve the confidence of results, an AdaRCCL algorithm is proposed. Consistency is ensured by ensuring that the predictions of base classifiers are similar, enabling the ensemble model to better capture patterns and structures in the data. This approach is also characterized by a reduced risk of overfitting, enhanced generalization capabilities on new data, and the mitigation of the negative impact of noise on classification results [43]. Simultaneously, the exclusion of diversity by consistency is avoided, and certain differences among base classifiers can be integrated, incorporating multiple perspectives in modeling the data. Specifically, the consistency among representation coefficients is achieved by the AdaRCCL algorithm, which assumes that different base classifiers in the ensemble learning should have relatively consistent behavior patterns for the same target sample. The distance between representation coefficients is described by an initial weight matrix in the objective function of the algorithm, thus improving the consistency among them. Ultimately, the more accurate and reliable base classifiers obtained using the AdaRCCL algorithm can be used to calculate SPCI, thereby improving the classification accuracy of testing samples and the overall classification performance of the system. The algorithm's detailed description and derivation are as follows.

A. Sample Boosting Based on Probabilistic Confidence Index

Give a training set $\boldsymbol{X}_{\text{tr}} = \{(\boldsymbol{x}_{\text{tr},i}, \hat{y}_{\text{tr},i})\}$, $i = 1, \ldots, N$; $\boldsymbol{x}_{\text{tr},i} \in \mathcal{R}^d$; $\hat{y}_{\text{tr},i} \in \mathcal{R}^d$; $|\mathcal{Y}| = K$, and a testing set $\boldsymbol{X}_{\text{te}} = \{(\boldsymbol{x}_{\text{te},j}, \hat{y}_{\text{te},j})\}$, $j = 1, \ldots, M$; $\boldsymbol{x}_{\text{te},j} \in \mathcal{R}^d$; $\hat{y}_{\text{te},j} \in \mathcal{R}^d$; $|\mathcal{Y}| = K$, which is learned from a set of base classifiers $\boldsymbol{F} = \{f_1, \ldots, f_Q\}$. Then, the problem is transformed to find an optimal method to integrate all these base classifiers to obtain an optimal classification result. These are common assumptions for ensemble learning approaches. According to (1)–(8), a straightforward assumption is that the reconstruction error can be used to describe the confidence of a testing sample with corresponding base classifier. That is, the lower the reconstruction error of a sample, the higher the confidence of the classification result. Then, the samples with high confidence can be selected to add training data and assist to classify the unlabeled samples with lower confidence. To accomplish the above goals, a novel sample boosting algorithm based on SPCI is proposed.

Definition 1 (Probabilistic Described Reconstruction Error): According to the assumption and analysis above, first, a probabilistic reconstruction error is defined as

$$\boldsymbol{\alpha}_{q,j}^{(0)} = \left(\kappa\left(\boldsymbol{X}_{\text{tr}}^{(0)}, \boldsymbol{X}_{\text{tr}}^{(0)}\right) + \lambda_1 \boldsymbol{I}\right)^{-1} \kappa(\boldsymbol{X}_{\text{tr}}, \boldsymbol{X}_{\text{te}}) \tag{9}$$

$$\boldsymbol{e}_{q,j}^{(0)} = \exp\left(-\left\|\boldsymbol{X}_{\text{te},j}^{(0)} - \boldsymbol{X}_{\text{tr}}^{(0)} \boldsymbol{\alpha}_{q,j}^{(0)}\right\|_2^2 / \left\|\boldsymbol{\alpha}_{q,j}^{(0)}\right\|_2^2\right) \tag{10}$$

where $\boldsymbol{\alpha}_{q,j}^{*,(0)}$ is the representation coefficient of testing samples $\boldsymbol{X}_{\text{te},j}^{(0)}$, which is learned from base classifier f_q, $\boldsymbol{X}_{\text{tr}}^{(0)}$ is the initial training data matrix, and $\boldsymbol{X}_{\text{te}}^{(0)}$ is the initial testing data matrix.

However, the confidence index is not reliable since the ensemble result from diverse base classifiers is used to calculate the reconstruction error. To address the problem that ensemble results with lower confidence, an approach that each of the base classifier has its own level of sample confidence is proposed. For a testing sample $\boldsymbol{x}_{\text{te},j}$, the output of each base classifier is given in a probability form, which is the probability that $\boldsymbol{x}_{\text{te},j}$ is correctly assigned to label by the $\boldsymbol{q}$th base classifier. Notably, the probability here is obtained by the co-calculation of multiple classifiers, which is given as

$$\boldsymbol{p}_{q,j}^{(0)} = \frac{\boldsymbol{e}_{q,j}^{(0)}}{\sum_{q=1}^{Q} \boldsymbol{e}_{q,j}^{(0)}} \tag{11}$$

$$\boldsymbol{P}^{(0)} = \begin{bmatrix} \boldsymbol{p}_{1,1}^{(0)} & \cdots & \boldsymbol{p}_{1,M}^{(0)} \\ \vdots & \ddots & \vdots \\ \boldsymbol{p}_{Q,1}^{(0)} & \cdots & \boldsymbol{p}_{Q,M}^{(0)} \end{bmatrix} \tag{12}$$

where $\boldsymbol{e}_{q,j}^{(0)}$ is the reconstruction error of testing sample $\boldsymbol{X}_{\text{te},j}^{(0)}$, learned from base classifier $\boldsymbol{f}_q$ classifier, $\boldsymbol{p}_{q,j}^{(0)}$ is a PRI of testing sample $\boldsymbol{X}_{\text{te},j}^{(0)}$, which is defined as reconstructed errors of the classifier pool $\boldsymbol{F}$, and $\boldsymbol{P}^{(0)}$ is the initial PRI matrix.

Definition 2 (Probabilistic Confidence Index Construction Based on Shannon Entropy): From (9)–(12), a probabilistic outputs of $\boldsymbol{Q}$ base classifiers on testing samples are obtained. Then, the problem is transformed to how to measure the probabilistic output confidence of the multiclassifier joint on a testing sample $\boldsymbol{X}_{\text{te},j}^{(0)}$. Since the probability output of each base classifier is obtained, the entropy can be used to measure certainty of the output. Therefore, a probabilistic confidence index by base classifier and Shannon entropy is defined as

$$\begin{aligned} \boldsymbol{se}_j^{(0)} &= \frac{1}{Q} \sum_{q=1}^{Q} \text{Shannon entropy}\left(\boldsymbol{p}_{q,j}^{(0)}\right) \\ &= \frac{1}{Q} \sum_{q=1}^{Q} -\boldsymbol{p}_{q,j}^{(0)} \log\left(\boldsymbol{p}_{q,j}^{(0)}\right) \quad \text{s.t. } \boldsymbol{p}_{q,j}^{(0)} > 0, \ \sum_{q=1}^{Q} \boldsymbol{p}_{q,j}^{(0)} = 1 \end{aligned} \tag{13}$$

$$\boldsymbol{SE}_j^{(0)} = \left[\boldsymbol{se}_1^{(0)}, \ldots, \boldsymbol{se}_M^{(0)}\right] \tag{14}$$

where $\boldsymbol{se}_{\mathbf{j}}^{(\mathbf{0})}$ is the confidence of testing sample $\boldsymbol{X}_{\boldsymbol{te},j}^{(\mathbf{0})}$, which is measured by the Shannon entropy of PRI for each base classifier $\boldsymbol{F}$, and $\boldsymbol{SE}_{\boldsymbol{j}}^{(\mathbf{0})}$ is the initial sample confidence matrix.

Definition 3 (Sample Boosting): According to (14), when the confidence index of $\boldsymbol{X}_{\text{te}}$ for each base classifier is acquired, the prediction of testing samples with high confidence index values is selected as candidate samples and added to initial training samples. Note that the smaller $\boldsymbol{SE}_j^{(0)}$ is, the higher

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certainty in the prediction of ensemble classifiers, and the higher confidence of testing sample $\boldsymbol{X}_{\text{te},j}^{(0)}$. Therefore, the samples with the top-$\boldsymbol{P}$ smallest entropy indices are chosen as

$$\boldsymbol{SE}_P^{(0)}\left(\boldsymbol{X}_{\text{te}}^{(0)}\right) = \text{smallest}_p(\boldsymbol{SE}) \tag{15}$$

$$\boldsymbol{X}_{\text{tr}}^{(1)} = \boldsymbol{X}_{\text{tr}}^{(0)} + \boldsymbol{SE}_P^{(0)}\left(\boldsymbol{X}_{\text{te}}^{(0)}\right) \tag{16}$$

$$\hat{y}_{\text{tr}} = \hat{y}_{\text{tr}}^{(0)} + \text{voting}\left(\boldsymbol{F}\left(\boldsymbol{X}_{\text{te}}^{(0)}, \boldsymbol{X}_{\text{te}}^{(0)}, \boldsymbol{\alpha}_q^{(0)}, \boldsymbol{SE}_P^{(0)}\left(\boldsymbol{X}_{\text{te}}^{(0)}\right)\right)\right) \tag{17}$$

where $\boldsymbol{SE}_P^{(0)}(\boldsymbol{X}_{\text{te}}^{(0)})$ is the first $\boldsymbol{P}$ samples with the highest confidence in the testing set $\boldsymbol{X}_{\text{te}}^{(0)}$, voting$(\boldsymbol{F}(\boldsymbol{X}_{\text{te}}^{(0)}, \boldsymbol{X}_{\text{te}}^{(0)}, \boldsymbol{\alpha}_q^{(0)}, \boldsymbol{SE}_P^{(0)}(\boldsymbol{X}_{\text{te}}^{(0)})))$ is the labels of the first $\boldsymbol{P}$ samples with the highest confidence in the testing set samples, which are voted by all the base classifiers $\boldsymbol{F}$, and $\{(\boldsymbol{X}_{\boldsymbol{tr}}^{(1)}, \hat{y}_{\text{tr}}^{(1)})\}$ is the amplified training sample and corresponding label learned from $\boldsymbol{F}$.

Algorithm Iteration:

From (9)–(17), an iterative SPCI algorithm is proposed. Update $\boldsymbol{X}_{\text{tr}}, \hat{y}_{\text{tr}}$: from iteration number $\boldsymbol{t} = 1$ to $\boldsymbol{t} = T$ or the convergence condition is satisfied the iterative algorithm updates training set $(\boldsymbol{X}_{\text{tr}}, \hat{y}_{\text{tr}})$ as

$$\boldsymbol{X}_{\text{tr}}^{(t+1)} = \boldsymbol{X}_{\text{tr}}^{(t)} + \boldsymbol{SE}_P^{(t)}\left(\boldsymbol{X}_{\text{te}}^{(t)}\right) \tag{18}$$

$$\hat{y}_{\text{tr}}^{(t+1)} = \hat{y}_{\text{tr}}^{(t)} + \text{voting}\left(F\left(\boldsymbol{X}_{\text{te}}^{(t)}, \boldsymbol{X}_{\text{te}}^{(t)}, \boldsymbol{\alpha}_q^{(t)}, \boldsymbol{SE}_P^{(t)}\left(\boldsymbol{X}_{\text{te}}^{(t)}\right)\right)\right) \tag{19}$$

where $(\boldsymbol{X}_{\text{tr}}^{(t+1)}, \hat{y}_{\text{tr}}^{(t+1)})$ is the final training set and corresponding labels obtained by ensemble of all base classifiers.

Update $\boldsymbol{\alpha}^*$: From iteration number $\boldsymbol{t} = 1$ to $\boldsymbol{t} = T$ or the convergence condition is satisfied, the iterative algorithm updates representation coefficient $\boldsymbol{\alpha}_q$ as

$$\boldsymbol{\alpha}_q^{(t+1)} = \left(\kappa\left(\boldsymbol{X}_{\text{tr}}^{(t+1)}, \boldsymbol{X}_{\text{tr}}^{(t+1)}\right) + \lambda_1'\mathbf{I}\right)^{-1}\kappa\left(\boldsymbol{X}_{\text{tr}}^{(t+1)}, \boldsymbol{X}_{\text{te}}^{(t+1)}\right) \tag{20}$$

where $\boldsymbol{\alpha}_q^{(t+1)}$ is the representation coefficient obtained by final training set, which is learned from the $\boldsymbol{q}$th base classifier.

In addition, the convergence condition of the iterative SPCI is that the size of new training samples is more than the size of testing samples.

B. Adaptive Representation Coefficient Consistency Learning

Although the accuracy of the base classifier can be effectively improved by SPCI through sample amplification and boosting, the base classifier itself is not improved. Therefore, the SPCI definition will not be reliable when the classification performance of initial base classifier is lower. To make the classification patterns between various base classifiers consistent and increase the confidence of the results, an AdaRCCL algorithm is proposed. The basic hypothesis of the proposed AdaRCCL is that for the same target sample, different base classifiers in ensemble learning should have relatively consistent behavior patterns. Obviously, the classification patterns of a CR-based classifier can be reflected by representation coefficient. It is natural to optimize $\|\bar{\boldsymbol{\alpha}}^{(0)} - \boldsymbol{\alpha}_q^{(0)}\boldsymbol{W}_q^{*,(0)}\|_2^2$ to make the classification patterns to be consistent. The algorithm details are described as follows.

First, according to (6), the representation coefficient of each classifier is given as $\boldsymbol{A} = \{\boldsymbol{\alpha}_1^{(0)}, \ldots, \boldsymbol{\alpha}_Q^{(0)}\}$, where $\boldsymbol{A}$ is the representation coefficient matrix that learns from $\boldsymbol{Q}$ KCRC base classifiers and $\boldsymbol{\alpha}_q^{(0)}$ is an initial representation coefficient that learns from the $\boldsymbol{q}$th base classifiers. Then, the joint classification pattern of $\boldsymbol{F}$ is described by the mean of the coefficients

$$\bar{\boldsymbol{\alpha}}^{(0)} = \frac{1}{Q}\sum_{q=1}^{Q}\boldsymbol{\alpha}_q^{(0)} \tag{21}$$

where $\bar{\boldsymbol{\alpha}}^{(0)}$ is the average representation coefficient calculated by all base classifiers. The objective function of our proposed AdaRCCL is described as

$$\boldsymbol{W}_q^{(0)} = \arg\min_{\boldsymbol{W}_q^{*,(0)}} \left\|\bar{\boldsymbol{\alpha}}^{(0)} - \boldsymbol{\alpha}_q^{(0)}\boldsymbol{W}_q^{*,(0)}\right\|_2^2 + \lambda_2\left\|\boldsymbol{W}_q^{*,(0)}\right\|_2^2$$

$$\Longrightarrow \boldsymbol{W}_q^{(0)} = \left(\boldsymbol{\alpha}_q^{(0)^T}\boldsymbol{\alpha}_q^{(0)} + \lambda_2\boldsymbol{I}\right)^{-1}\boldsymbol{\alpha}_q^{(0)^T}\bar{\boldsymbol{\alpha}}^{(0)} \tag{22}$$

where the initial weight matrix $\boldsymbol{W}_q^{*,(0)}$ is used to represent the distance between $\bar{\boldsymbol{\alpha}}^{(0)}$ and $\boldsymbol{\alpha}_q^{(0)}$. The weights are then assigned back to the representation coefficients, and a novel weighted kernel CR algorithm is proposed as follows:

$$\boldsymbol{\alpha}_q^{(1)} = \arg\min_{\boldsymbol{\alpha}_q^*} \|\boldsymbol{\Phi}(\boldsymbol{X}_{\text{te}}) - \boldsymbol{\Phi}(\boldsymbol{X}_{\text{tr}})\|_2^2 + \lambda_3\left\|\boldsymbol{W}_q^{(0)}\boldsymbol{\alpha}_q^*\right\|_2^2$$

$$\Longrightarrow \boldsymbol{\alpha}_q^{(1)} = \left(\kappa(\boldsymbol{X}_{\text{tr}}, \boldsymbol{X}_{\text{tr}}) + \lambda_3\boldsymbol{W}_q^{(0)^T}\boldsymbol{W}_q^{(0)}\right)^{-1}\kappa(\boldsymbol{X}_{\text{tr}}, \boldsymbol{X}_{\text{te}}) \tag{23}$$

where $\boldsymbol{\alpha}_q^{(1)}$ is the new representation coefficient that learns from the $\boldsymbol{q}$th W-KCRC base classifiers with the weight matrix $\boldsymbol{W}_q^{(0)}$ is considered.

Algorithm Iteration: From (21)–(23), an iterative AdaRCCL is proposed.

Update $\bar{\boldsymbol{\alpha}}$: From iteration number $\boldsymbol{t} = 1$ to $\boldsymbol{t} = T$ or the convergence condition is satisfied, the iterative algorithm updates representation coefficient $\bar{\boldsymbol{\alpha}}$ as

$$\bar{\boldsymbol{\alpha}}^{(t)} = \frac{1}{Q}\sum_{q=1}^{Q}\boldsymbol{\alpha}_q^{(t)}. \tag{24}$$

Update W: From iteration number $\boldsymbol{t} = 1$ to $\boldsymbol{t} = T$ or the convergence condition is satisfied, the iterative algorithm updates representation coefficient $\boldsymbol{W}_{\boldsymbol{q}}$ as

$$\boldsymbol{W}_q^{(t)} = \left(\boldsymbol{\alpha}_q^{(t)^T}\boldsymbol{\alpha}_q^{(t)} + \lambda_2\boldsymbol{I}\right)^{-1}\boldsymbol{\alpha}_q^{(t)^T}\bar{\boldsymbol{\alpha}}^{(t)}. \tag{25}$$

Update $\boldsymbol{\alpha}_q$: From iteration number $\boldsymbol{t} = 1$ to $\boldsymbol{t} = T$ or the convergence condition is satisfied, the iterative algorithm updates representation coefficient $\boldsymbol{\alpha}_{\boldsymbol{q}}$ as

$$\boldsymbol{\alpha}_q^{(t+1)} = \left(\boldsymbol{X}_{\text{tr}}^T\boldsymbol{X}_{\text{tr}} + \lambda_3\boldsymbol{W}_q^{(t)^T}\boldsymbol{W}_q^{(t)}\right)^{-1}\boldsymbol{X}_{\text{tr}}^T\boldsymbol{X}_{\text{te}}. \tag{26}$$

The final representation coefficient is used to calculate the residual and the classification results according to (7) and (8).

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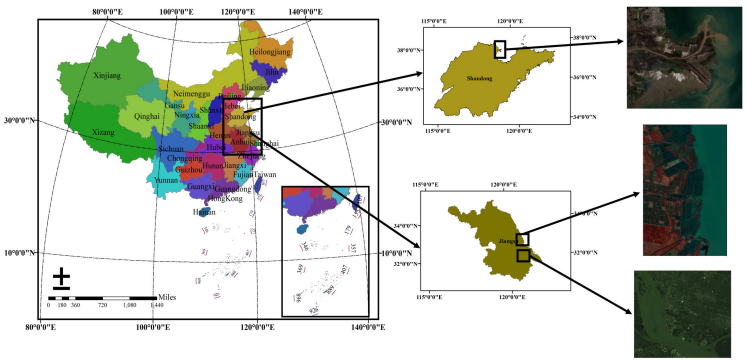


Fig. 2. Location maps of the study area.

In addition, the convergence condition of the iterative AdaRCCL is designed as

$$\begin{cases} \text{if } \boldsymbol{s}^{(t+1)} >= \boldsymbol{s}^{(t)}, & \text{Convergence} \\ \text{otherwise}, & \text{Not convergence}. \end{cases} \tag{27}$$

$\boldsymbol{s}^{(t+1)}$ is given in

$$\boldsymbol{s}^{(t+1)} = \left\| \bar{\boldsymbol{\alpha}}^{(t+1)} - \boldsymbol{\alpha}_q^{(t+1)} \boldsymbol{W}_q^{(t)} \right\|_2^2 \tag{28}$$

where $\|\bar{\boldsymbol{\alpha}}^{(t+1)} - \boldsymbol{\alpha}_q^{(t+1)} \boldsymbol{W}_q^{(t)}\|_2^2$ is used to measure the distance between the representation coefficient of each base classifier and the mean of the representation coefficients.

C. Novel ProESS Framework

Since AdaRCCL is proposed to help improve the reliability of sample amplification in SPCI, it is natural to build a new ProESS framework based on SPCI and AdaRCCL. To reduce the computational complexity, only one-time sample amplification was used to replace the iterative sample amplification process in SPCI. In addition, parameters $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_1'$, $\boldsymbol{\lambda}_2$, and $\boldsymbol{\lambda}_3$ in ProESS all use the optimal parameters in SPCI and AdaRCCL and use different values of parameter $\boldsymbol{P}$ for experiments. The specific steps of ProESS are shown in Fig. 1 and pseudocode 3.

IV. Experiments

In this article, three GF-5 Chinese coastland wetland hyperspectral data sets are selected for classification experiments.

A. GF-5 Satellite and Data Preprocessing

The GF-5 satellite was launched on May 9, 2018, at the Taiyuan Satellite Launch Center in China. It is the only hyperspectral satellite among the seven civilian satellites of the high-resolution earth observation system major project. It is designed as a sun-synchronous orbit with an orbital altitude of about 705 km. The GF-5 satellite is the first satellite that simultaneously conducts comprehensive observations of land and the atmosphere in the world. With a width of 60 km and a spatial resolution of 30 m, 330 spectral color channels can be obtained from visible to short-wave infrared (0.4–0.25 μm) spectral range, and the spectral coverage is nearly 9 times wider than that of ordinary cameras. The number of bands is nearly a hundred of times that of ordinary cameras, and its spectral resolution in the visible bands is 5 nm. Users can submit requests to China Landsat Data Center (http://data.cresda.com:90/#/home) and High-Resolution Earth Observation System Grid Platform (https://www.cheosgrid.cn/app/search/search.htm) to apply for data download. Currently, only protocol users can download GF-5 data [50]. In this article, the GF-5 images of the Yangtze River Delta (Nantong) region and Jiangsu Dafeng Natural Reserve on July 9, 2019, and the GF-5 images of the Yellow River Delta region on January 7, 2019, were used for wetland classification. The locations of our study areas are shown in Fig. 2. Data preprocessing was performed using ENVI software. Sample selection according to discontinuity, quantitative adequacy, and typicality was performed using a combination of field sampling and visual interpretation. For hyperspectral remote sensing data, ENVI software is often used for data preprocessing, such as radiometric correction and atmospheric correction. The main parameters of radiation correction and atmospheric correction are set as follows: the sensor altitude is 705 km, and the pixel size is 30 m; the atmospheric model and aerosol model are set according to data center latitude and acquisition time, and other parameters are default values; the reference images are Landsat 8 satellite images, and the DEM is Aster GDEM V2 data.

TABLE I

Detailed Description of the Sample in Yangtze River Delta (Nantong) Dataset

Class	Ground object	Number
1	river	1273
2	tree	575
3	bare land (bal)	200
4	farmland (far)	189
5	lake	137
6	mud flat (muf)	476
7	flood plain (flp)	435
8	impervious surface (ims)	1046
9	plant	455
Total		4786

B. Wetland Hyperspectral Datasets

The Yangtze River Delta (Nantong) (31° 40′–31° 55′ N, 20° 45′–121° 10′ E) of Jiangsu Province was used as the first wetland data to verify the effectiveness of the proposed

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TABLE II
DETAILED DESCRIPTION OF THE SAMPLE IN DAFENG NATURAL RESERVE DATASET

Class	Ground object	Number
1	sea	1020
2	mud flat (muf)	528
3	road	150
4	building (bui)	89
5	farmland (far)	918
6	river	276
7	tree	724
8	aquaculture (acq)	746
9	lake	79
Total		4530

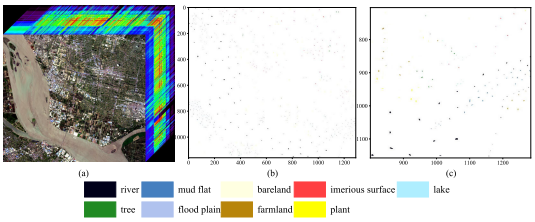


Fig. 3. (a) False-color image. (b) Ground truth. (c) Magnified ground truth of Yangtze River Delta (Nantong) dataset.

algorithm. As located near a city with a relatively high degree of urbanization, the wetland is greatly affected by human activities. These data contain 268 spectral bands (after removing 62 bad bands) with wavelengths ranging from 0.39 to 2.5 μm [51], [52]. Its spatial size is 1160 $\times$ 1209. The details of nine classes in this HSI are described in Table I, and the images are shown in Fig. 3. The second wetland data selected for the experiment are Dafeng Natural Reserve (32° 56′–33° 36′ N, 120° 42′–120° 51′ E), acquired on the sea-shore of the Yellow Sea. Different from the first wetland data, these data are located in a nature reserve, and the ground features are mainly natural biological plants, which are less affected by human beings. This image contains 256 spectral bands (after removing 74 bad bands) with wavelengths ranging from 0.39 to 2.5 μm. It contains 986 $\times$ 632 pixels, and the spatial resolution is 30 m. The descriptions of classes in this data are listed in Table II, and images are shown in Fig. 4. The Yellow River Delta dataset (37° 15′–38° 10′ N, 118° 10′–119° 15′ E) with 1185 $\times$ 1324 pixels is used as the third data to evaluate the proposed methods. Yellow River Delta is located at the estuary of the Yellow River in Huanghe Kou town, Kenli district, Dongying, Shangdong province. The wavelengths of spectral bands ranging from 0.4 to 2.5 μm. Two hundred eighty-five effective bands remained in the data after removing 45 bad bands. The image includes 21 classes, and the spatial resolution is 30 m. The detailed information of each class is described in Table III, and images of this dataset are shown in Fig. 5.

C. Experimental Setup

All experiments are implemented on the platform of Python 3.8.6. The specific parameters of the proposed algorithms are set as follows. The number of initial samples $N = 10$, the number of selected samples $P = \{200, 400, 600, 800, 1000\}$, regularization parameter for KCRC $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_1' = \{$1e-1, 1e-2, 1e-3, 1e-4, 1e-5$\}$, regularization parameter for W-KCRC $\boldsymbol{\lambda}_2 = \{$1e-1, 1e-2, 1e-3, 1e-4, 1e-5$\}$, and regularization parameter for weighted calculated $\boldsymbol{\lambda}_3 = \{$1e-1, 1e-2, 1e-3, 1e-4, 1e-5$\}$.

TABLE III
DETAILED DESCRIPTION OF THE SAMPLE IN YELLOW RIVER DELTA DATASET

Class	Ground object	Number
1	acquaculture (acq)	393
2	seep sea (ses)	796
3	soybean (soy)	110
4	rice	190
5	building (bui)	83
6	maize	96
7	broomcorn (bro)	95
8	locust	211
9	spartina (spa)	200
10	shallow sea(shs)	936
11	mud flat (muf)	553
12	river	469
13	suaeda salsa (sus)	361
14	reed	240
15	salt marsh (sam)	595
16	intertidal saltwater (ins)	454
17	tamarix (tam)	133
18	pond	377
19	flood plain (flp)	68
20	freshwater herbaceous marsh (fhm)	72
21	aquatic vegetation (aqv)	39
Total		6491

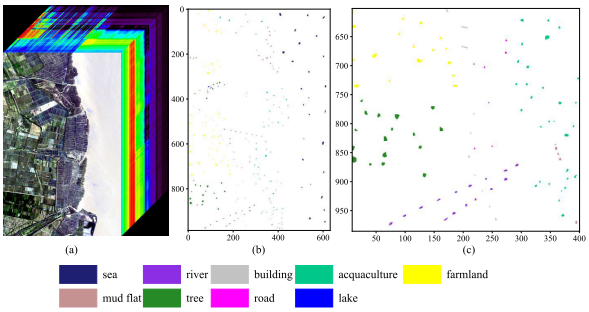


Fig. 4. (a) False-color image. (b) Ground truth. (c) Magnified ground truth of Dafeng Natural Reserve dataset.

D. Baseline and Comparative Algorithms

Based on the two main aspects of our article, namely, the proposed SPCI for sample augmentation and AdaRCCL for sample quality improvement, it is deemed inappropriate to compare the proposed method with only a single classifier. Therefore, the following experiments were conducted to demonstrate the superiority of the proposed algorithm. First, the existing support vector machine (SVM) [53] and random forest (RF) [54] classifiers are augmented by incorporating label propagation (LP) as a sample augmentation technique, resulting in LP-SVM and LP-RF classifiers. This allows for the comparison of the performance of our proposed technique against LP-based methods. Second, the tri-training algorithm is combined with Catboost [55], LightGBM [56], and gradient boosting decision tree (GBDT) [56] classifiers to improve

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TABLE IV

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR YANGTZE RIVER DELTA (NANTONG) (TEN TRAINING SAMPLES WERE RANDOMLY SELECTED FOR EACH CLASS; LP IS LABEL PROPAGATION ALGORITHM; TRI1 IS TRI-TRAINING BASED ON CATBOOST, LIGHTGBM, AND GBDT; TRI2 IS TRI-TRAINING WITH CR-BASED CLASSIFIERS)

class	Train	SVM	RF	LP-SVM	LP-RF	Tri 1	Tr 2	GCN	GCN-A	LightGBM	Meta-DES	**SPCI**	**AdaRCCL**	**ProESS**
1	10	92.24	93.82	93.19	89.55	95.01	96.04	96.28	92.32	86.94	93.82	92.64	92.64	92.16
2	10	64.25	60.71	50.80	46.02	19.82	72.57	60.71	16.46	72.39	70.62	54.34	66.90	61.77
3	10	87.37	86.32	84.74	88.42	100.00	85.79	100.00	91.58	85.79	87.37	88.42	95.79	95.26
4	10	96.67	98.33	85.00	98.33	98.33	92.78	95.00	95.56	92.78	100.00	95.00	95.56	95.56
5	10	99.21	76.38	38.58	82.68	95.28	83.46	77.17	87.40	81.10	85.83	83.46	94.49	88.98
6	10	53.86	76.61	63.30	74.89	83.69	55.58	74.46	88.41	51.29	64.81	75.97	78.11	82.83
7	10	68.24	93.65	95.76	91.76	70.59	94.35	91.29	90.12	91.06	92.00	99.76	95.76	96.47
8	10	57.72	61.78	71.04	68.34	9.94	69.02	18.34	77.41	66.99	59.27	65.44	90.44	92.95
9	10	90.34	93.48	96.18	93.48	100.00	98.88	85.39	84.04	99.10	90.34	98.43	94.38	96.18
OA		75.26	80.41	78.62	78.86	82.48	64.48	70.75	78.50	78.79	79.80	81.24	88.35	**88.69**
AA		78.88	82.34	75.40	81.50	83.16	74.74	77.63	80.37	80.83	82.67	83.72	**89.34**	89.13
Kappa		0.71	0.77	0.75	0.75	0.79	0.59	0.66	0.75	0.75	0.76	0.78	0.86	**0.87**

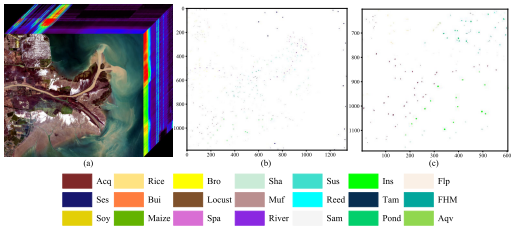


Fig. 5. (a) False-color image. (b) Ground truth. (c) Magnified ground truth of Yellow River Delta dataset.

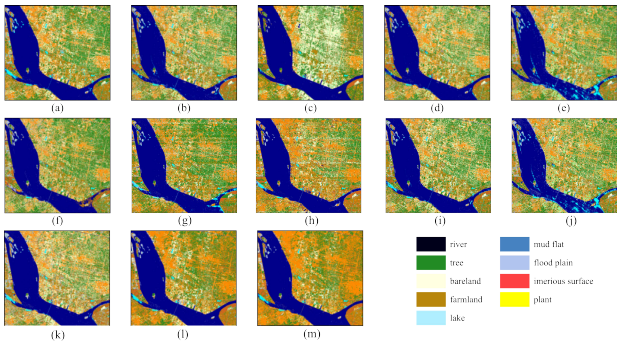


Fig. 6. Classification maps obtained by different models of Yangtze River Delta (Nantong). (a) SVM. (b) RF. (c) LP-SVM. (d) LP-RF. (e) Tr 1. (f) Tr 2. (g) GCN. (h) GCN-A. (i) LightGBM. (j) Meta-DES. (k) SPCI. (l) AdaRCCL. (m) ProESS.

sample quality and augment the dataset, in contrast to our proposed ProESS method. This allows for the comparison of the performance of our proposed algorithm with the state-of-the-art boosting methods. Furthermore, the tri-training algorithm is also combined with the CR-based classifiers used in our proposed algorithm to demonstrate the superior performance of our proposed method in semi-supervised learning. Finally, two state-of-the-art deep learning algorithms were added as comparison algorithms, namely, deep graph convolutional network (GCN) [57] and deep attention graph convolutional (GCN-A) [58]. In addition, since the method of this article is based on the idea of dynamic ES, the additional two ensemble learning methods are added as comparison algorithms, namely, LightGBM [59] and Meta-DES [60]. Overall, these modifications enable a more comprehensive evaluation of the effectiveness of our proposed algorithm in both sample augmentation and quality improvement, as well as its ability to perform well in semi-supervised learning.

E. Classification Performance and Maps

Tables IV–VI show the accuracy of three wetland datasets under the same training sample size. The overall accuracy (OA), average accuracy (AA), and kappa indexes are used to evaluate the classification of all the comparisons and proposed methods in this section. From Table IV, for the Yangtze River Delta (Nantong), the best classification performance 88.69% is obtained by ProESS, which improved by 6.21% than Tri-training 1. The other two proposed methods are also superior to the comparison algorithms. It can be seen from Table V that the highest classification accuracy 93.42% of Dafeng Natural Reserve is also acquired from ProESS. Compared with the classic ensemble learning methods LP-SVM and LP-RF, the accuracy yields 8.46% and 7.27% improvements. Moreover, the three proposed algorithms have excellent performance compared to the three well-known ensemble methods. Like the other two wetland datasets, ProESS still has the highest accuracy for Yellow River Delta 95.31%, improved by 2.68% and 4.73% than the SPCI and AdaRCCL, respectively.

Additionally, the classification accuracy of AdaRCCL algorithm for three data is better than SPCI algorithm and slightly lower than ProESS. Based on the above observations, it is obvious that the three proposed methods can obtain higher classification results than the comparison algorithms. The optimal experimental results of ProESS show the necessity of combining SPCI and AdaRCCL. The classification maps generated by the proposed methods and other comparative algorithms are shown in Figs. 6–8. For Yangtze River Delta (Nantong), from the experiment results from Table IV and Fig. 6, it can be seen that SPCI has obtained the highest accuracy for flood plain and also has a better performance for other ground objects. AdaRCCL has the best classification results for bare land. Notably, even the lowest accuracy for the tree in AdaRCCL is much better than the other comparison algorithms. Moreover, ProESS performance classification results for all the ground objects in Yangtze River Delta (Nantong). The highest classification performance of 82.83% and 92.95%

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TABLE V

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR DAFENG NATURAL RESERVE (TEN TRAINING SAMPLES WERE RANDOMLY SELECTED FOR EACH CLASS; LP IS LABEL PROPAGATION ALGORITHM; TRI1 IS TRI-TRAINING BASED ON CATBOOST, LIGHTGBM, AND GBDT; TRI2 IS TRI-TRAINING WITH CR-BASED CLASSIFIERS)

class	Train	SVM	RF	LP-SVM	LP-RF	Tri 1	Tri 2	GCN	GCN-A	LightGBM	Meta-DES	**SPCI**	**AdaRCCL**	**ProESS**
1	10	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	98.02	100.00	100.00	100.00	100.00
2	10	63.71	78.19	80.89	78.19	89.00	83.01	71.04	57.53	85.14	83.01	87.45	95.17	99.23
3	10	75.00	77.14	59.29	67.14	17.86	68.57	66.43	85.00	77.14	77.14	80.71	87.14	87.14
4	10	100.00	100.00	100.00	92.50	100.00	100.00	100.00	98.75	78.75	100.00	100.00	97.50	100.00
5	10	76.43	77.42	77.20	78.08	86.67	85.02	76.32	87.56	76.76	75.44	85.57	87.33	88.11
6	10	47.74	69.92	25.19	59.77	46.24	84.21	72.18	82.33	85.34	71.43	69.92	68.80	68.42
7	10	95.38	95.52	97.20	96.36	99.72	96.08	94.40	95.52	96.64	95.38	95.52	94.68	96.36
8	10	89.54	87.50	93.21	86.55	87.91	77.72	80.71	89.67	74.05	88.04	88.45	93.61	93.89
9	10	75.36	84.06	47.83	72.46	86.96	85.51	85.51	95.65	94.20	84.06	73.91	89.86	89.86
OA		84.17	87.28	84.96	86.15	88.47	87.93	84.73	88.45	86.15	87.59	90.14	92.46	**93.42**
AA		80.35	85.53	75.64	81.23	86.68	79.37	82.95	88.00	85.12	86.06	86.84	90.46	**91.44**
Kappa		0.81	0.85	0.82	0.83	0.86	0.85	0.82	0.86	0.84	0.85	0.88	0.91	**0.92**

TABLE VI

CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR YELLOW RIVER DELTA (TEN TRAINING SAMPLES WERE RANDOMLY SELECTED FOR EACH CLASS; LP IS LABEL PROPAGATION ALGORITHM; TRI1 IS TRI-TRAINING BASED ON CATBOOST, LIGHTGBM, AND GBDT; TRI2 IS TRI-TRAINING WITH CR-BASED CLASSIFIERS)

class	Train	SVM	RF	LP-SVM	LP-RF	Tri 1	Tri 2	GCN	GCN-A	LightGBM	Meta-DES	**SPCI**	**AdaRCCL**	**ProESS**
1	10	91.64	89.56	85.64	89.03	80.16	91.38	87.47	86.95	86.42	90.86	92.17	79.63	87.99
2	10	100.00	98.98	100.00	98.09	100.00	99.11	100.00	100.00	98.73	99.11	100.00	100.00	100.00
3	10	99.01	95.05	99.01	98.02	96.04	95.05	97.03	90.10	98.02	99.01	98.02	99.01	99.01
4	10	88.33	87.22	88.33	88.89	88.33	72.22	87.78	88.33	79.44	87.78	88.89	88.33	88.33
5	10	35.62	69.86	73.97	69.86	41.10	69.86	42.47	82.19	64.38	58.90	68.49	86.30	86.30
6	10	98.84	91.86	100.00	91.86	100.00	90.70	100.00	100.00	88.37	91.86	100.00	100.00	100.00
7	10	97.67	97.67	97.67	98.84	100.00	96.51	98.84	100.00	93.02	98.84	98.84	100.00	100.00
8	10	100.00	99.01	100.00	100.00	100.00	95.05	100.00	98.51	95.05	99.01	100.00	100.00	100.00
9	10	99.47	100.00	99.47	100.00	100.00	100.00	100.00	100.00	97.89	100.00	100.00	100.00	100.00
10	10	45.57	87.90	99.03	93.52	100.00	91.36	99.24	87.80	98.27	90.39	97.95	95.57	99.14
11	10	97.05	96.69	97.05	95.76	97.24	96.87	95.40	99.82	92.08	96.87	96.87	100.00	100.00
12	10	100.00	100.00	100.00	100.00	100.00	100.00	87.58	100.00	99.13	100.00	100.00	100.00	100.00
13	10	91.45	98.01	96.01	98.29	100.00	86.61	70.37	98.01	82.05	98.29	98.58	96.58	96.58
14	10	71.30	74.78	71.30	70.43	13.91	79.57	13.91	53.91	76.52	76.52	62.17	71.74	73.48
15	10	100.00	100.00	100.00	99.83	100.00	100.00	98.80	98.46	98.80	100.00	100.00	100.00	100.00
16	10	98.87	84.91	98.42	83.78	94.82	93.02	93.69	99.10	86.49	84.91	95.27	89.64	89.19
17	10	78.05	86.18	78.05	76.42	75.61	78.86	80.49	93.50	82.93	85.37	86.99	86.99	89.43
18	10	60.76	77.93	61.04	77.93	43.32	67.30	16.08	77.11	68.12	77.11	80.38	85.56	81.74
19	10	98.28	77.59	98.28	86.21	100.00	82.76	100.00	100.00	67.24	81.03	93.10	96.55	96.55
20	10	82.54	84.13	82.54	87.30	23.81	92.06	92.06	93.65	58.73	84.13	90.48	93.65	93.65
21	10	96.55	96.55	96.55	96.55	96.55	82.76	96.55	96.55	89.66	96.55	96.55	96.55	96.55
OA		85.47	92.13	91.54	90.51	91.60	89.35	85.95	93.07	90.58	92.63	94.83	94.41	**95.31**
AA		87.19	90.18	93.71	92.56	88.62	83.38	83.70	92.57	85.78	90.31	92.61	93.62	**94.19**
Kappa		0.84	0.91	0.93	0.92	0.91	0.88	0.85	0.92	0.90	0.92	0.94	0.94	**0.95**

for mud flat and imperious surface is obtained by ProESS. From Table V and Fig. 7, for Dafeng Natural Reserve, SPCI method can get the best classification performance for sea and building, while AdaRCCL obtains the highest classification accuracy for sea and road. Like Yangtze River Delta (Nantong), ProESS has better classification performance for almost all the ground objects in Dafeng Natural Reserve, except for rivers, trees, and lakes. The three proposed algorithms have excellent performance for all ground objects for the last Yellow River Delta wetland dataset. The best classification results for aquaculture and rice are also shown in ProESS. Moreover, it should be noted that AdaRCCL and ProESS have almost the same trend in the classification accuracy of all the ground objects, especially for the classification results of ground objects building, locust, and mud flat, which is much better than other algorithms.

V. DISCUSSION

A. Parameter Sensitivity Analysis

1) Number of Selected Samples P and Regularization Parameter $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_1'$ for SPCI: From Figs. 9–11, the 3-D plots of three wetland datasets are used to demonstrate the impact of using different values of $\boldsymbol{P}$, $\boldsymbol{\lambda}_1$, and $\boldsymbol{\lambda}_1'$. Fig. 9(a)–(e) shows the effect of different values for parameter $\boldsymbol{\lambda}_1$, on accuracy in Yangtze River Delta (Nantong). For each subfigure, the x-axis presents the five different values for $\boldsymbol{P}$, while the y-axis shows the five different values for $\boldsymbol{\lambda}_1'$. The corresponding classification accuracy of SPCI is shown in the z-axis. It can be seen that the color trend of the surface plotted does not change much along the y-axis, that is, the parameter $\boldsymbol{\lambda}_1'$ has little impact on the classification accuracy. On the contrary, the parameter $\boldsymbol{P}$ has a great influence on the classification

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Algorithm 1 SPCI

Input: the initial training set $\boldsymbol{X}_{tr} \in \mathcal{R}^{d\times N}$, $\boldsymbol{X}_{te} \in \mathcal{R}^{d\times M}$, **and** the base classifier generate by bootstrap $\boldsymbol{F} = \{f_1, \ldots, f_Q\}$

a. Get the initial representation coefficient $\boldsymbol{\alpha}_{q,j}^{(0)}$ of each base classifier by **Eq.** (9)
b. Construct the SPCI model by **Eq.** (10)-(17)
c. Repeat the following steps until convergence

Step 1 Update $\boldsymbol{X}_{tr}, \hat{y}_{tr}$ by **Eq.** (18) and **Eq.** (19)
Step 2 Update $\boldsymbol{\alpha}^*$ by **Eq.** (20)

d. Check convergence condition number of new training samples is more than the testing samples size

Output: New training samples $\boldsymbol{X}_{tr}^{(t+1)}$ is used for classification

Algorithm 2 AdaRCCL

Input: the initial training set $\boldsymbol{X}_{tr} \in \mathcal{R}^{d\times N}$, $\boldsymbol{X}_{te} \in \mathcal{R}^{d\times M}$, and the base classifier generate by bootstrap $\boldsymbol{F} = \{f_1, \ldots, f_Q\}$

a. Get the initial representation coefficient $\boldsymbol{\alpha}_{q,j}^{(0)}$ of each base classifier by **Eq.** (9)
b. Construct the AdaRCCL model by **Eq.** (21)-(24)
c. Repeat the following steps until convergence

Step 1 update $\overline{\boldsymbol{\alpha}}$ by **Eq.** (25)
Step 2 Update $\boldsymbol{W_q}$**:** by **Eq.** (26)
Step 3 Update $\boldsymbol{\alpha_q}$ by **Eq.** (27)
Check convergence condition

$$\begin{cases} if\ \boldsymbol{s}^{(t+1)} >= \boldsymbol{s}^{(t)}, & Convergence \\ otherwise, & Not\ convergence \end{cases}$$

Output: New base classifier reconstructed by $\boldsymbol{W_q}$.

Algorithm 3 ProESS

Input: the initial training set $\boldsymbol{X}_{tr} \in \mathcal{R}^{d\times N}$, $\boldsymbol{X}_{te} \in \mathcal{R}^{d\times M}$, **and** the base classifier generate by bootstrap $\boldsymbol{F} = \{f_1, \ldots, f_Q\}$

a. Get the initial representation coefficient $\boldsymbol{\alpha}_{q,j}^{(0)}$ of each base classifier by Eq. (9)
b. Generate new training samples $\boldsymbol{X}_{tr}^{(2)}$ by algorithm 1 (only one-time of iteration)
c. Use the new training samples $\boldsymbol{X}_{tr}^{(2)}$ as input to build algorithm 2

Output: New training samples $\boldsymbol{X}_{tr}^{*,(2)}$ is used for classification

accuracy, and the surface color plotted changes significantly along the x-axis.

For Jiangsu Dafeng Natural Reserve and Yellow River Delta, similar conclusions can be obtained from Figs. 10 and 11. In general, for three wetland datasets, the effect of parameter $\boldsymbol{P}$ on classification accuracy is larger than the other two parameters $\boldsymbol{\lambda}_1$, and $\boldsymbol{\lambda}_1'$. However, the gap of the accuracy for the three wetland datasets with different $\boldsymbol{P}$, $\boldsymbol{\lambda}_1$, and $\boldsymbol{\lambda}_1'$ values is insignificant. It indicates that the proposed SPCI algorithm is very robust to its parameters. Therefore, the SPCI algorithm can very easily obtain high classification

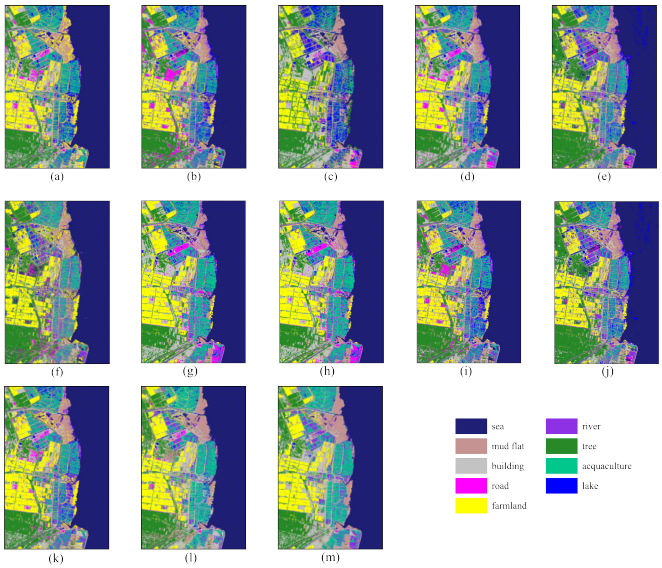


Fig. 7. Classification maps obtained by different models of Dafeng Natural Reserve. (a) SVM. (b) RF. (c) LP-SVM. (d) LP-RF. (e) Tr 1. (f) Tr 2. (g) GCN. (h) GCN-A. (i) LightGBM. (j) Meta-DES. (k) SPCI. (l) AdaRCCL. (m) ProESS.

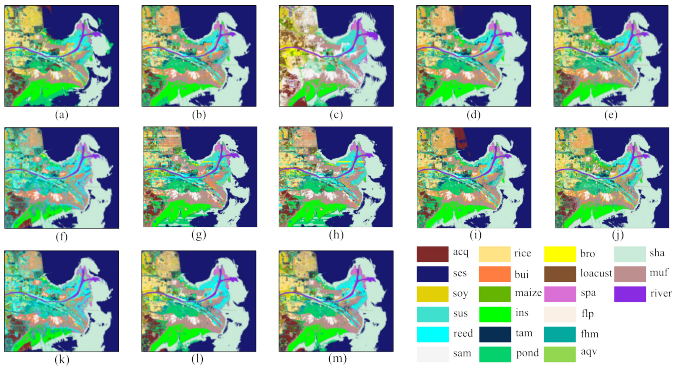


Fig. 8. Classification maps obtained by different models of Yellow River Delta. (a) SVM. (b) RF. (c) LP-SVM. (d) LP-RF. (e) Tr 1. (f) Tr 2. (g) GCN. (h) GCN-A. (i) LightGBM; (j) Meta-DES. (k) SPCI. (l) AdaRCCL. (m) ProESS.

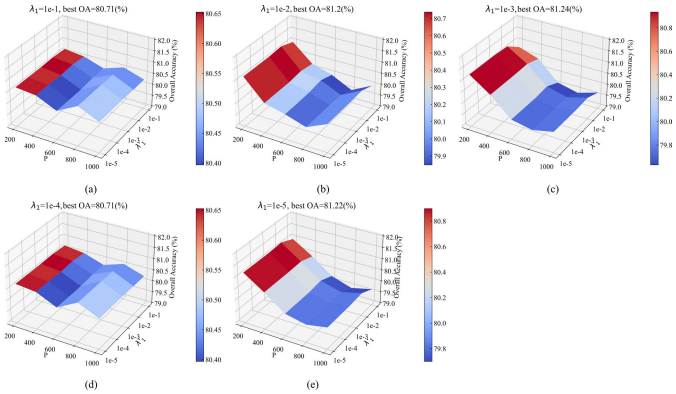


Fig. 9. Effect of parameter $\boldsymbol{P}$, $\boldsymbol{\lambda}_1$, and $\boldsymbol{\lambda}_1'$ on OA for SPCI, with ten samples of each class in Yangtze River Delta (Nantong). (a) $\boldsymbol{\lambda}_1$ = 1e-1. (b) $\boldsymbol{\lambda}_1$ = 1e-2. (c) $\boldsymbol{\lambda}_1$ = 1e-3. (d) $\boldsymbol{\lambda}_1$ = 1e-4. (e) $\boldsymbol{\lambda}_1$ = 1e-5.

performance in real-world wetland classification problems and does not require much effort on tuning parameters.

It is worth noting that the impact of increasing the $\boldsymbol{P}$ value on classification accuracy varies across different datasets. Specifically, in the Nantong dataset, the OA decreases with

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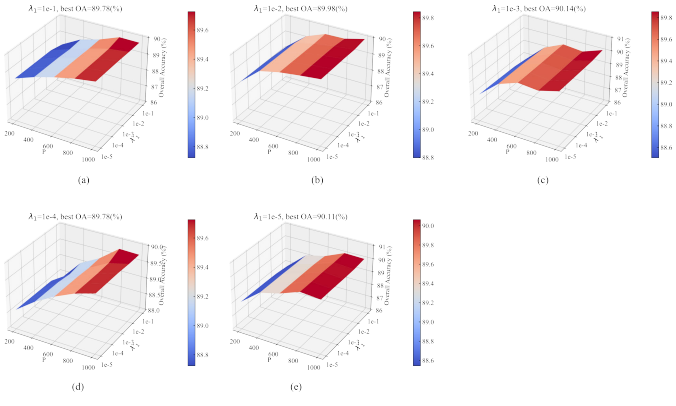


Fig. 10. Effect of parameter $\boldsymbol{P}$, $\boldsymbol{\lambda}_1$, and $\boldsymbol{\lambda}_1'$ on OA for SPCI, with ten samples of each class in Jiangsu Dafeng Natural Reserve. (a) $\boldsymbol{\lambda}_1$ = 1e-1. (b) $\boldsymbol{\lambda}_1$ = 1e-2. (c) $\boldsymbol{\lambda}_1$ = 1e-3. (d) $\boldsymbol{\lambda}_1$ = 1e-4. (e) $\boldsymbol{\lambda}_1$ = 1e-5.

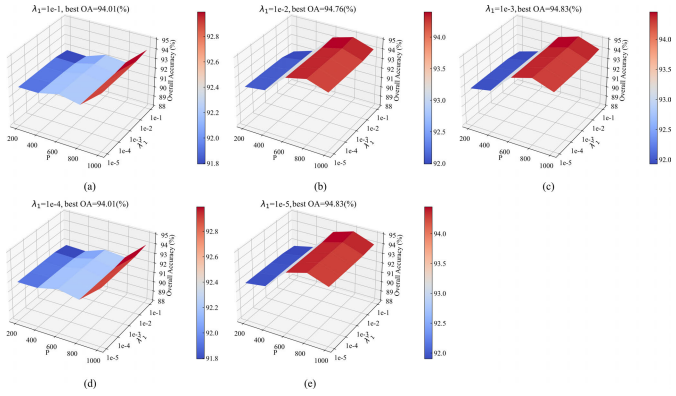


Fig. 11. Effect of parameter $\boldsymbol{P}$, $\boldsymbol{\lambda}_1$, and $\boldsymbol{\lambda}_1'$ on OA for SPCI, with ten samples of each class in Yellow River Delta. (a) $\boldsymbol{\lambda}_1$ = 1e-1. (b) $\boldsymbol{\lambda}_1$ = 1e-2. (c) $\boldsymbol{\lambda}_1$ = 1e-3. (d) $\boldsymbol{\lambda}_1$ = 1e-4. (e) $\boldsymbol{\lambda}_1$ = 1e-5.

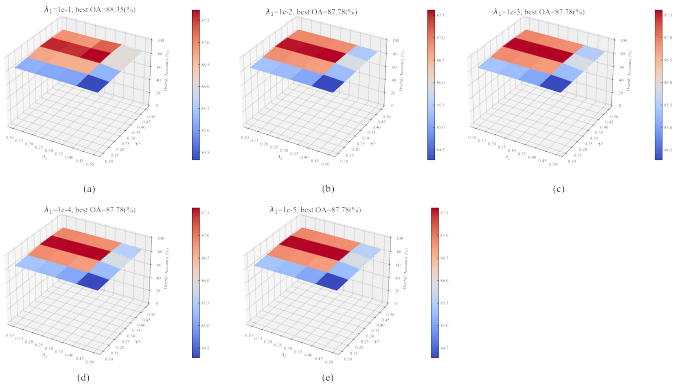


Fig. 12. Effect of parameter $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$, and $\boldsymbol{\lambda}_3$ on OA for AdaRCCL, with ten samples of each class in Yangtze River Delta (Nantong) dataset. (a) $\boldsymbol{\lambda}_1$ = 1e-1. (b) $\boldsymbol{\lambda}_1$ = 1e-2. (c) $\boldsymbol{\lambda}_1$ = 1e-3. (d) $\boldsymbol{\lambda}_1$ = 1e-4. (e) $\boldsymbol{\lambda}_1$ = 1e-5.

the increase of $\boldsymbol{P}$ (as shown in Fig. 9), while in other datasets, the OA increases with the increase of $\boldsymbol{P}$ (as shown in Figs. 10 and 11). The reason for the decrease of classification accuracy when the $\boldsymbol{P}$ value reaches a certain level in the Nantong dataset may be due to overfitting caused by too many samples, which reduces the model's generalization ability. In addition, factors, such as differences between datasets, land cover types, spatial resolution, and spectral response, may lead to changes in classification accuracy. Therefore, to achieve the best classification results in different datasets, it is necessary to

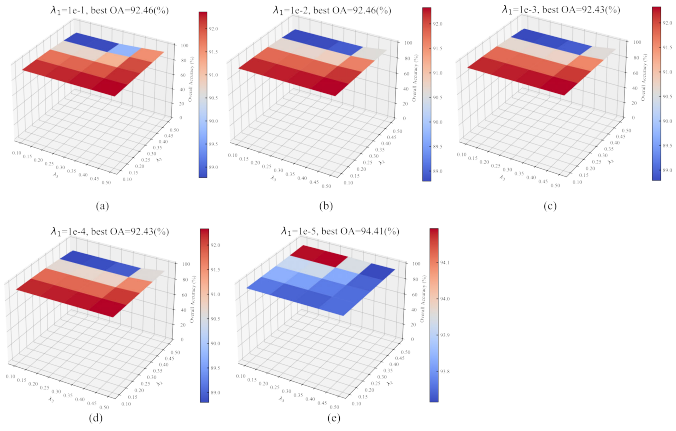


Fig. 13. Effect of parameter $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$, and $\boldsymbol{\lambda}_3$ on OA for AdaRCCL, with ten samples of each class in Jiangsu Dafeng Natural Reserve. (a) $\boldsymbol{\lambda}_1$ = 1e-1. (b) $\boldsymbol{\lambda}_1$ = 1e-2. (c) $\boldsymbol{\lambda}_1$ = 1e-3. (d) $\boldsymbol{\lambda}_1$ = 1e-4. (e) $\boldsymbol{\lambda}_1$ = 1e-5.

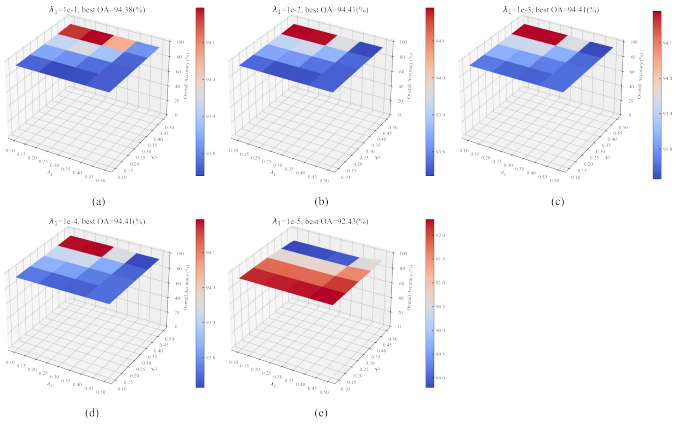


Fig. 14. Effect of parameter $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$, and $\boldsymbol{\lambda}_3$ on OA for AdaRCCL, with ten samples of each class in Yellow River Delta. (a) $\boldsymbol{\lambda}_1$ = 1e-1. (b) $\boldsymbol{\lambda}_1$ = 1e-2. (c) $\boldsymbol{\lambda}_1$ = 1e-3. (d) $\boldsymbol{\lambda}_1$ = 1e-4. (e) $\boldsymbol{\lambda}_1$ = 1e-5.

carefully consider the selection of a suitable $\mathbf{P}$ value to avoid the risks of overfitting or underfitting.

2) Regularization Parameter $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$, and $\boldsymbol{\lambda}_3$ for AdaRCCL: Similar to the algorithm SPCI, three 3-D plots are used to present the effect on OA by using different values of $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$, and $\boldsymbol{\lambda}_3$ (see Figs. 12–14). For each subfigure, it can be seen that all surfaces are almost flat, and the accuracy of the three-wetland data does not change much as the parameter $\boldsymbol{\lambda}_1$ changes. This observation indicates that the proposed AdaRCCL algorithm is not sensitive to its parameters. Therefore, the AdaRCCL is a robust wetland classification algorithm that can be conveniently adopted for real application.

3) Number of Selected Samples P for ProESS: Since the optimal regularization parameters obtained from SPCI and AdaRCCL are used in the ProESS, only the effect of parameter $\boldsymbol{P}$ of ProESS is discussed in this section. In Fig. 15, a line chart of classification performance in the three wetland datasets is used to illustrate the impact of using different values of $\boldsymbol{P}$. For each value [200, 400, 600, 800, 1000] for $\boldsymbol{P}$ in x-axis, the accuracy of corresponding is shown in the y-axis.

From Fig. 15, it is shown that the accuracy trend of all three-wetland datasets increases first and then decreases with the increase of $\boldsymbol{P}$ values. In a specific analysis of the classification performance of Yangtze River Delta (Nantong)

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TABLE VII
ABLATION STUDY ON THE WETLAND DATASETS FOR SAMPLE AUGMENTATION

Algorithm Description						
	Tri-training (CR-based Classifiers) compare with **SPCI**	√				
	Tri-training (Catboost, LightGBM, GBDT) compare with **AdaRCCL**		√			
	SPCI (CR-based Classifiers)			√		√
	AdaRCCL				√	√
Three HSI Datasets	Yangtze River Delta (OA%)	64.68	82.29	**81.24 (+16.56)**	**88.35 (+23.67)**	**88.69 (+24.01)**
	Jiangsu Dafeng Natural Reserve (OA%)	87.93	88.18	**90.14 (+2.21)**	**92.46 (+4.53)**	**93.42 (+5.49)**
	Yellow River Delta (OA%)	89.35	91.48	**94.83 (+5.48)**	**94.41 (+5.06)**	**95.31 (+5.96)**

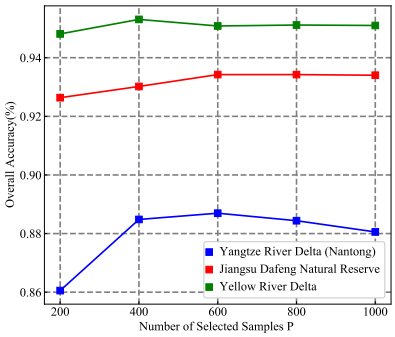


Fig. 15. Effect of parameter ***P*** on OA for ProESS, with ten samples of each class in three wetland datasets.

dataset, the highest accuracy is obtained when the ***P*** value is 200. The classification accuracy of Jiangsu Dafeng Natural Reserve and Yellow River Delta datasets reaches the optimal when the ***P*** value is 600. However, the accuracy gap for three wetland datasets with different ***P*** values is from 0.5% to 3%, which is insignificant. This observation demonstrates that the proposed ProESS is robust to the parameter ***P***. Therefore, the ProESS can be very easy and stable to obtain reliable results in real-world wetland classification problems.

B. Ablation Study

Based on the experimental results presented in Table VII, it can be observed that higher accuracy is exhibited by SPCI and AdaRCCL methods than Tri-training and classic sample boosting methods (such as Catboost, LightGBM, and GBDT) across all three wetland hyperspectral datasets. Notably, on the Yellow River Delta dataset, significantly higher accuracy is achieved by SPCI and AdaRCCL methods than the other methods. From a sample augmentation perspective, the number of training samples can be effectively increased by the SPCI method, which can help improve the robustness and generalization performance of classifiers. It can be found that classifier accuracy can be significantly improved by sample augmentation, particularly for tasks with limited training samples, by comparing the results of tri-training and SPCI methods. The accuracy can be improved by approximately 16% on the Yangtze River Delta and Jiangsu Dafeng Natural Reserve datasets and approximately 5% on the Yellow River Delta dataset by the SPCI method. From a sample quality perspective, key samples in the training data can be enhanced and the influence of noise and outliers can be reduced by the AdaRCCL method, thereby improving classifier accuracy and stability. Across all three datasets, the AdaRCCL method exhibits higher accuracy than classic sample boosting methods, such as Catboost, LightGBM, and GBDT. The accuracy can be improved by approximately 7% on the Yangtze River Delta dataset, approximately 2% on the Jiangsu Dafeng Natural Reserve dataset, and approximately 5% on the Yellow River Delta dataset by the AdaRCCL method.

C. SPCI-Based Sample Amplification Effectiveness Analysis

In Table VIII, the accuracy of the proposed SPCI algorithm and other semi-supervised learning methods on three wetland datasets is compared. The table shows the accuracy of the different methods on different numbers of amplified samples (400, 500, and 600 samples). The comparison is made between the LP, tri-training based on CatBoost, LightGBM, and GBDT (Tri1), tri-training with CR-based classifiers (Tri2), voting, and the proposed SPCI algorithm. From the results, it can be observed that the proposed SPCI-based method consistently outperformed the other semi-supervised learning methods across all three wetland datasets and at different numbers of amplified samples. For instance, in the Yangtze River Delta dataset, the accuracy of the SPCI algorithm was 0.90 with 400 amplified samples, which is higher than the accuracy of all other methods. In the Jiangsu Dafeng Natural Reserve dataset, the accuracy of the SPCI algorithm was 0.93 with 600 amplified samples, which is again higher than the accuracy of all other methods. Similarly, in the Yellow River Delta dataset, the accuracy of the SPCI algorithm was 0.94 with 400 amplified samples, which is higher than the accuracy of all other methods. These results demonstrate the effectiveness, superiority, and robustness of the proposed SPCI algorithm relative to other algorithms in the context of sample amplification and semi-supervised learning. In conclusion, the data presented in Table VIII provide strong evidence to support the effectiveness and innovation of the proposed SPCI algorithm in improving the accuracy of semi-supervised learning across different numbers of amplified samples and different wetland datasets.

D. Ensemble Effectiveness Analysis

For the ensemble learning algorithm, a key point is whether the final ensemble result is better than the accuracy of each

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TABLE VIII

AMPLIFIED SAMPLES ACCURACY OF THE PROPOSED SPCI ALGORITHM VERSUS OTHER SEMI-SUPERVISED METHODS FOR THREE WETLAND DATASETS. (LP IS LABEL PROPAGATION ALGORITHM; TRI1 IS TRI-TRAINING BASED ON CATBOOST, LIGHTGBM, AND GBDT; TRI2 IS TRI-TRAINING WITH CR-BASED CLASSIFIERS; VOTING IS THE RESULT OF MAJORITY VOTE)

Sample Amplification	400 samples					500 samples					600 samples				
	LP	Tri1	Tri2	Voting	**PSCI**	LP	Tri1	Tri2	Voting	**PSCI**	LP	Tri1	Tri2	Voting	**PSCI**
Yangtze River Delta (OA)	0.80	0.42	0.85	0.68	**0.90**	0.79	0.43	0.84	0.67	**0.92**	0.77	0.41	0.82	0.62	**0.93**
Jiangsu Dafeng Natural Reserve (OA)	0.83	0.84	0.79	0.68	**0.91**	0.80	0.81	0.76	0.70	**0.92**	0.80	0.87	0.75	0.69	**0,93**
Yellow River Delta (OA)	0.89	0.82	0.89	0.83	**0.94**	0.90	0.81	0.90	0.80	**0.93**	0.91	0.83	0.91	0.83	**0.94**

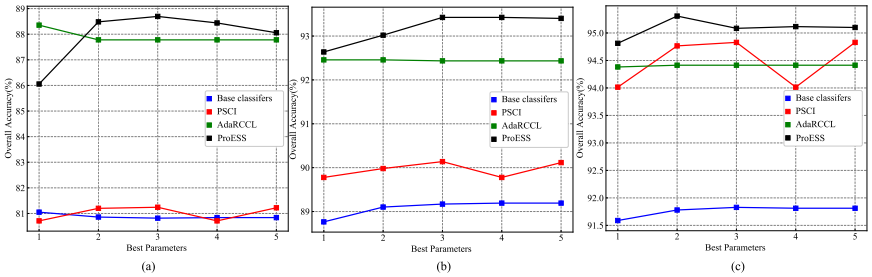


Fig. 16. Effect of parameter $\boldsymbol{\lambda}_1$, $\boldsymbol{\lambda}_2$, and $\boldsymbol{\lambda}_3$ on OA for AdaRCCL, with ten samples of each class in Yellow River Delta.

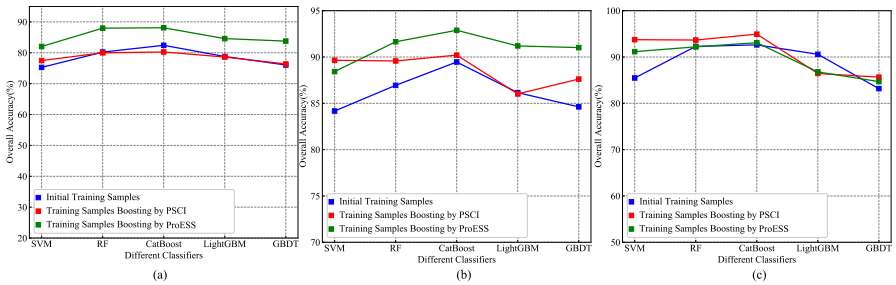


Fig. 17. Applicability of amplifying samples from AdaRCCL and ProESS in different machine learning algorithms, with ten samples of each class in three wetland datasets.

base classifier. Thus, additional experiments are used to validate the ensemble effectiveness of our methods. Fig. 16 shows the accuracy of three proposed algorithms and base classifiers. From Fig. 16(b) and (c), it can be seen that both methods are significantly better than the results of base classifier. This demonstrates the ensemble effectiveness of the proposed algorithm on Jiangsu Dafeng Natural Reserve and Yellow River Delta. Notably, from Fig. 16(a), the classification performance by SPCI is very similar to the base classifier. However, the classification accuracy is much higher than that of the base classifier when using the AdaRCCL and ProESS.

E. Algorithm Extensibility Analysis

Here, the extensibility of the proposed AdaRCCL and ProESS is discussed based on several machine learning methods. Essentially, both AdaRCCL and ProESS can be regarded as a method of sample amplification. Therefore, it is necessary to evaluate whether the final training samples obtained by them are equally effective in other classifier algorithms. Fig. 17 illustrates the classification performance with initial training samples compared with the results of new training samples, which AdaRCCL and ProESS acquire. For each machine learning algorithm in the x-axis, the accuracy of corresponding is shown in the y-axis. From the results of Fig. 17(a) and (b), it can be seen that for Yangtze River Delta (Nantong) and Jiangsu Dafeng Natural Reserve, AdaRCCL and ProESS can effectively improve the classification accuracy for all the algorithms. Especially for Jiangsu Dafeng Natural Reserve, the proposed sample amplification algorithm can improve the classification accuracy by nearly 5%. From the experiments of Fig. 17(c), although the classification accuracy of the LightGBM algorithm is not improved using new training samples, for several other classifier models, AdaRCCL and ProESS are still effective for Yellow River Delta. Thus, in real-world problems, the proposed methods can be used as sample amplification steps, combined with other more state-of-the-art classification algorithms to solve the small sample problem in wetland hyperspectral classification.

F. Algorithm Transferability Analysis

The Botswana wetland dataset was acquired by NASA EO-1 satellite in Botswana's Okavango Delta in May 2001. The

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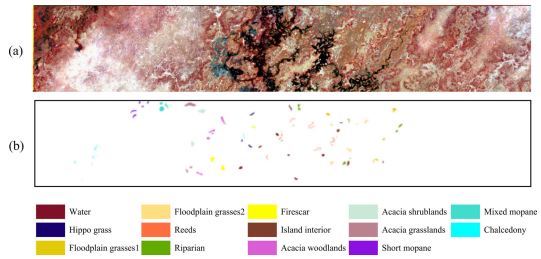


Fig. 18. (a) False-color image. (b) Ground truth of Botswana dataset.

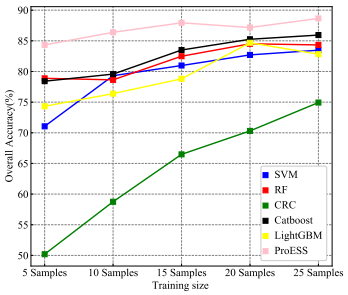


Fig. 19. Accuracy of ProESS versus different machine learning algorithms, with different samples of each class in Botswanan dataset.

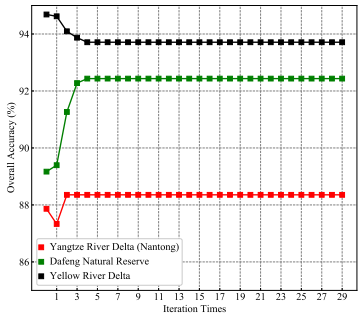


Fig. 20. Effect of iteration times $\boldsymbol{T}$ on OA for AdaRCCL, with ten samples of each class in three wetland datasets.

image size is 1476 × 256 (Fig. 18). The wavelength range of the sensor on EO-1 is 0.4–0.25 μm, and the spatial resolution is about 20 m. In 242 bands, noise bands (1–9, 56–81, 98–101, 120–133, and 165–186) are removed, and 145 bands are used for training.

There are 14 types of ground objects in the study area. It can be seen from Fig. 19 that the proposed ProESS algorithm can offer better classification performance compared with other algorithms under different training sample sizes. It indicates that the proposed algorithm still has excellent transferability on EO-1 satellite data. In addition, it should be pointed out that with the increase in the number of samples, the classification accuracy of ProESS algorithm does not change much. In addition, Fig. 19 shows that the classification accuracy of ProESS with small samples is higher than most comparison algorithms with large samples. Overall, the transferability and effectiveness of ProESS are proved based on the above observation.

G. Extensive Analysis Based on Multisources HSI Datasets

Two HSI datasets from different sources were added as new experimental data in order to validate the effectiveness of the proposed method on data from different sensors. The first dataset is hyperspectral remote sensing data from the Purdue University (Purdue Campus) Lafayette campus. The data were collected on September 30, 1999, using an airborne hyperspectral mapper (HYMAP) sensor. The second dataset is the Houston HSI, acquired using the ITRES CASI-1500 sensor in and around Houston, TX, USA. The data have a spatial resolution of 2.5 m and a size of 349 × 1905 pixels. It consists of 144 spectral bands with a wavelength range of 364–1046 nm. The study area encompasses 15 land cover types, including roads, soil, trees, highways, and more. From Tables IX and X, it can be seen that for HSI data from different data sources, the three proposed methods still have the highest accuracy.

TABLE IX
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR PURDUE

	GCN	GCN-A	LightGBM	Meta-DES	SPCI	AdaRCCL	ProESS
OA	83.63	91.07	80.76	84.54	93.22	94.84	94.93
AA	83.81	93.21	84.82	87.61	94.04	94.09	93.83

TABLE X
CLASSIFICATION ACCURACY OF DIFFERENT METHODS FOR HOUSTON

	GCN	GCN-A	LightGBM	Meta-DES	SPCI	AdaRCCL	ProESS
OA	68.50	73.06	82.45	83.42	83.92	82.35	82.37
AA	70.23	75.37	82.45	83.74	84.26	84.43	84.49

H. Convergence Analysis

For SPCI and ProESS, it is obvious that when the times of iterations are large enough, the number of sample amplification reaches the size of the testing set, and the algorithm is converged. Therefore, the convergence of AdaRCCL is shown in Fig. 20. The AdaRCCL achieves convergence in all three datasets within ten iterations. It indicates that the convergence of the algorithm can be guaranteed. Notably, although the final classification performance is better than other algorithms for the Yellow River Delta dataset, the accuracy decreases as the number of iterations increases. For the other two wetland datasets, as the number of iterations increases, the classification accuracy increases until convergence. Overall, the AdaRCCL has good convergence capability on wetland datasets.

I. Computational Complexity

The computational complexity of the proposed methods is analyzed in this section. Because the SPCI and AdaRCCL focus on the acquisition of representation coefficients and weighted coefficients, only the computational complexity about the two parts is given. First, in (9), the complexity is $O(N^3 + N^2d + \text{NMd})$. For the step in (20), when the number of iterations is T, the complexity is $O((N+\text{TP})^3 + (N+\text{TP})^2d + (N+\text{TP})\text{Md})$. Second, in (23), the complexity is $O(M^3 + M^2N)$, and when the number of iterations is $\boldsymbol{T}$, the computational complexity of (23) is $O(M^3 + M^2(N+\text{TP}))$.

VI. CONCLUSION

This study proposed a novel probabilistic ES framework to solve the problems of small sample sizes and low sample

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confidence in HSI for wetland classification. The SPCI is a sample ES method that can be divided into three subprocesses. First, the output of each base classifier is probabilistic from the representation coefficient; then, the sample confidence based on a single classifier is obtained. Second, the joint probability output from ensemble learning is obtained by combining multiple classifiers, and SPCI is defined. Finally, new training samples are obtained by SPCI to improve the classification accuracy.

However, the confidence of samples updated by SPCI is not enough by the low accuracy of base classifier, which will affect the improvement of classification accuracy. With the assumption that each base classifier in the ensemble should have relatively consistent behavior patterns for a testing sample, an AdaRCCL algorithm is proposed to solve this problem. This method satisfies the above hypothesis by defining a weight matrix to minimize the difference between the base classifier representation coefficient and the average coefficient. By constantly updating the weights and representation coefficients until the algorithm converges, the final representation coefficients are used to improve the classification performance. Finally, based on the SPCI and AdaRCCL, a novel ProESS framework is proposed to complete both sample amplification and boosting. To demonstrate the performance of the SPCI, AdaRCCL, and ProESS methods, a set of classic machine learning and state-of-the-art ensemble algorithms are compared. The experiments show that not only all the proposed algorithms can achieve higher accuracy in wetland classification, but they are also not sensitive to parameters and have high stability. On the other hand, to evaluate the ensemble effectiveness, the majority voting result of the base classifiers is used to compare with the proposed algorithms. Experimental results show that the classification performance of the SPCI, AdaRCCL, and ProESS is higher than the ensemble results of base classification, which shows that the ensemble is effective. Moreover, since the three proposed algorithms are essentially sample amplification methods, using amplified or initial samples are compared. The experimental results show that the selected and improved samples can improve the accuracy of most classification algorithms, indicating that the proposed algorithms have great expansion ability. Finally, the transferability performance is discussed by a new set of wetland hyperspectral data, and the results show that the proposed algorithm can produce good accuracy for different data. Experiments and analysis demonstrated that the proposed algorithms have excellent classification performance, extensibility, and transferability. The SPCI, AdaRCCL, and ProESS are robust to parameters setting and have excellent ensemble performance. Accurate wetland classification results can provide reliable information for wetland spatial distribution statistics, wetland area calculation, and wetland change monitoring, thus providing a solid foundation for wetland management and protection. However, it seems that AdaRCCL and ProESS do not provide optimal solutions when they converge. The reason for this could be that increasing the number of sample amplification does not necessarily result in higher classification accuracy. The more samples are amplified, the more erroneous samples may be introduced, leading to a decrease in classification accuracy. Therefore, in future research, it could be considered to treat the number of sample amplifications as a parameter and use intelligent optimization algorithms to obtain the optimal solution.

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